INTERPRETABLE MULTICLASS MODELING OF FORENSIC FIREARM DECISIONS IN THE BULLET BLACK BOX STUDY

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1 Abstract

This paper studies examiner decision outcomes from the Bullet Black Box Study using interpretable machine learning methods. We model the final examiner conclusion as a three-class problem: *Match*, *NonMatch*, or *Inconclusive*. Because the dataset contains two different comparison settings, Known-Questioned (KQ) and Questioned-Questioned (QQ), we analyze them separately and train separate models for each. We first build baseline models using Random Forest and then evaluate XGBoost as a second tree-based method. The Random Forest models achieve balanced accuracies of 0.681 for KQ and 0.691 for QQ, showing that the available features contain useful predictive signal. The XGBoost model performs especially well on the KQ dataset, reaching a test accuracy of 0.8077, while the QQ XGBoost model achieves a test accuracy of 0.6916. In addition to predictive performance, we focus on interpretation. Confusion matrices show that the most common difficulty is separating *Inconclusive* from *NonMatch*, which reflects the uncertainty present in many firearm comparisons. Feature importance results show that both technical bullet-mark variables and comparison-context variables, such as difficulty and quality measures, influence examiner decisions. Overall, the results suggest that machine learning can help identify meaningful patterns in firearm examination data while also highlighting the limits of prediction in cases where the evidence is ambiguous.

2 Introduction

Forensic firearm examination involves comparing bullets or cartridge cases to determine whether they were fired from the same firearm. These decisions are important because they can influence criminal investigations and courtroom testimony. In practice, examiners may reach one of three conclusions: *Match*, *NonMatch*, or *Inconclusive*. The inconclusive category is particularly important because it reflects situations where the available evidence does not support a clear identification or exclusion.

In this study, we analyze examiner decision outcomes from the public *Bullet Black Box Study*. Our goal is to understand whether these decisions can be predicted using technical comparison features and contextual information about the examination process. Instead of focusing only on predictive accuracy, we also aim to understand which factors are associated with different decision outcomes. To do this, we use interpretable machine learning methods that allow us to examine how individual variables influence model predictions.

The dataset contains two types of comparison tasks. In Known–Questioned (KQ) comparisons, a crime-scene bullet is compared to test-fired bullets from a recovered firearm. In Questioned–Questioned (QQ) comparisons, two unknown bullets are compared to determine whether they were fired from the same firearm. Because these comparison settings represent different forensic situations and contain slightly different variables, we analyze them separately and train independent models for each dataset.

Our analysis follows several steps. First, we prepare and explore the datasets to understand their structure and missing-value patterns. Next, we train baseline predictive models using Random Forests and evaluate an additional model using XGBoost. Finally, we analyze model behavior using confusion matrices and feature importance measures to understand which variables influence predictions.

The goal of this work is not to replace forensic examiners with automated systems. Instead, we aim to use machine learning as a tool for studying decision patterns in firearm comparison data. By examining which features are associated

with *Match*, *NonMatch*, and *Inconclusive* outcomes, we hope to better understand the conditions under which firearm comparisons become difficult and when uncertainty arises in examiner judgments.

3 Literature Review

Forensic firearm examination is a feature-comparison discipline in which examiners compare bullets or cartridge cases in order to determine whether they were fired from the same firearm, from different firearms, or whether the available evidence is insufficient to reach a clear conclusion [1, 2]. These decisions are important because they can influence criminal investigations and courtroom testimony. As a result, researchers and policy groups have increasingly emphasized the need for empirical studies that evaluate how reliable these comparison decisions are in practice. One widely used approach for studying forensic decision-making is the use of black-box studies. In these studies, examiners are asked to evaluate comparison items with known ground truth while working under conditions that resemble real forensic casework. The goal is to measure how often examiners make correct identifications or exclusions, and how frequently they report inconclusive results [1, 5]. Black-box studies are particularly valuable because they evaluate the full decision process rather than only testing isolated technical procedures.

The Bullet Black Box Study is one of the most comprehensive studies conducted on firearm examination. The study was designed to evaluate the accuracy and reproducibility of bullet comparison decisions across a large number of examiners and comparison scenarios. Importantly, the study includes challenging but realistic conditions such as damaged bullets, multiple firearm types, and both known-questioned and questioned-questioned comparison settings [1, 2]. Because of the scale and design of the study, it provides a rich dataset that allows researchers to analyze patterns in examiner decisions. Results from the Bullet Black Box Study and subsequent analyses by the NIST working group show that examiner performance cannot be fully summarized by a single overall error rate [2]. Instead, decision outcomes vary depending on multiple factors, including the type of firearm, ammunition characteristics, and the specific comparison scenario being evaluated. These findings highlight the importance of studying firearm examination outcomes at a more detailed level rather than relying only on aggregated statistics.

Another important topic in the literature concerns the interpretation of inconclusive decisions. Several researchers argue that inconclusive outcomes should not be treated as missing data or statistical noise [3, 4]. Instead, they represent meaningful situations in which the available evidence does not support a confident identification or exclusion. The way inconclusive responses are handled can significantly affect how error rates are reported and interpreted in forensic studies [3]. Because of this, many scholars recommend analyzing firearm comparison outcomes as a multiclass decision problem rather than reducing the analysis to a simple same-gun versus different-gun classification.

Prior work also shows that firearm comparison outcomes are influenced by several different sources of variability. Some variability comes from the physical characteristics of the evidence itself. For example, certain firearm types produce less distinctive markings on bullets, which can make comparisons more difficult. The Bullet Black Box reports highlight this issue in particular for bullets fired from firearms with polygonal rifling, such as Glock pistols. These bullets often produce fewer clear markings and therefore lead to higher rates of inconclusive decisions [2]. Other variability may arise from differences in examiner experience, interpretation of comparison features, or the overall difficulty of the comparison task. These findings suggest that firearm examination outcomes depend on a combination of evidence characteristics, comparison difficulty, and examiner behavior. As a result, studying these decisions requires methods that can capture relationships between many interacting variables rather than focusing on only one or two factors.

Machine learning methods provide useful tools for analyzing these types of datasets. The Bullet Black Box data consist of structured tabular data with multiple predictor variables describing bullet characteristics, comparison conditions, and contextual information. Models such as random forests are well suited to this type of data because they can capture nonlinear relationships and interactions between variables while remaining relatively robust to noise and missing values [6]. For this reason, random forests are often used as a strong baseline for tabular classification problems. However, predictive performance alone is not sufficient in forensic contexts. If a model predicts that a comparison is likely to result in a Match, NonMatch, or Inconclusive decision, it is also important to understand why that prediction was made. This requirement has motivated the development of interpretable machine-learning methods that allow researchers to analyze how model predictions are generated.

Two commonly used interpretability techniques are permutation feature importance and SHAP explanations. Permutation importance measures how much each variable contributes to predictive performance by observing how the model's accuracy changes when the values of that variable are randomly shuffled [8]. SHAP explanations provide a more detailed view by estimating how individual features contribute to the prediction for a specific case [7]. Together, these methods allow researchers to identify which variables are most influential in the model and how those variables affect individual predictions. Another important consideration when evaluating machine-learning models is probability calibration. A model may achieve good classification accuracy while still producing poorly calibrated probability

estimates. In other words, the predicted probabilities may not correspond well to the true likelihood of each outcome. In forensic applications, this distinction is important because overly confident predictions may give the impression of certainty even when the evidence is ambiguous [9]. Calibration analysis therefore provides an additional way to evaluate how well a model reflects uncertainty in its predictions.

Most previous research on firearm examination has focused on measuring examiner error rates and understanding how inconclusive decisions affect those estimates [1, 2, 3, 5]. In contrast, fewer studies have attempted to model the examiner decision process itself using interpretable machine-learning approaches. Our project builds on this literature by combining insights from forensic reliability studies with modern machine-learning methods. Specifically, we model examiner decisions as a multiclass prediction problem with three possible outcomes: Match, NonMatch, and Inconclusive. We train separate models for Known–Questioned and Questioned–Questioned comparisons and evaluate them using both predictive performance metrics and interpretability tools. The goal is not to replace the examiner with an automated system, but rather to better understand which comparison features and contextual factors are associated with different decision outcomes. By studying these patterns, we aim to provide additional insight into the conditions under which firearm comparisons become difficult and uncertainty arises in examiner judgments.

4 Data and Exploratory Analysis

Our analysis uses the public workbook `bullets_2024.xlsx`, which contains examiner response sheets together with comparison-set metadata from the Bullet Black Box Study. From this workbook, we created two model-ready datasets: `KQ_model_ready.csv` for Known-Questioned comparisons and `QQ_model_ready.csv` for Questioned-Questioned comparisons. We treat these as two separate datasets rather than combining them into a single table. The reason is that the two comparison settings represent different forensic situations and do not contain exactly the same variables. For example, some fields such as `Comparability` appear only in the KQ responses, while other variables are defined differently between the two settings. By keeping the datasets separate, we avoid forcing incompatible structures into one table and allow each comparison task to be analyzed within its own context.

The data-cleaning process was designed with the goal of predicting the examiner's final decision while keeping the structure of the original study intact. First, we standardized variable names across the response sheets so that the datasets would be easier to analyze consistently. We then mapped the raw examiner conclusions into three modeling classes: Match, NonMatch, and Inconclusive. After this step, each response table was merged with the comparison-set metadata using the `Cset` identifier. This merge step is important because the response sheets alone do not include enough information for modeling. Many useful predictors, such as bullet quality labels, assignment counts, intervening-bullet ranges, caliber information, and detailed comparison categories, are stored in the metadata sheet.

Our cleaning decisions were guided by two main principles. The first principle was to remove variables that would not be appropriate predictors. For example, examiner identifiers and similar fields were excluded because they identify the participant rather than the comparison itself. We also avoided including ground-truth style variables that could leak the correct answer into the model and artificially inflate predictive performance. The goal was to ensure that the predictors reflect information that would realistically be available during a comparison decision.

The second principle was to preserve as much of the original data as possible. Instead of removing all rows with missing values, we examined whether the missingness was structural or accidental. In several cases the missing values arise from how the study was designed. For instance, in the KQ dataset the second-questioned-bullet quality variables are completely missing because KQ comparisons involve only one questioned bullet. Similarly, the intervening-bullets variables are missing for many rows in both datasets because that information is not defined for every comparison set. Because these patterns reflect the design of the study rather than data errors, we kept the rows and treated the missing values as part of the dataset structure to be handled later during preprocessing.

After these preparation steps, the final model-ready datasets contain 1,575 rows and 23 columns for the KQ data and 1,581 rows and 22 columns for the QQ data. Each row represents a single examiner comparison outcome along with technical bullet features, quality assessments, and contextual information about the comparison set. These datasets provide the foundation for our multiclass prediction models while remaining faithful to the structure and meaning of the original Bullet Black Box Study data.

4.1 Target Variable and Feature Structure

The target variable in both datasets is the examiner's final conclusion, grouped into three classes:

- **Match**: `ID`, `LeanID`
- **NonMatch**: `Excl`, `LeanExcl`

- **Inconclusive**: `Insuff`, `NoValue`

This mapping is important because *Inconclusive* is treated as a meaningful forensic outcome rather than as missing data or a discarded case. That choice makes the prediction problem more realistic and better aligned with actual firearm-examination practice, where ambiguity and insufficient evidence are legitimate endpoints.

The predictor variables fall into several broad groups: decision-context variables such as `Difficulty`, `Comparability`, and `Phase`; bullet-quality variables from prescreen and voted assessments; firearm and bullet descriptors such as caliber and detailed comparison category; and contextual variables such as assignment counts and intervening-bullet ranges. These features were retained because they may help explain not only why a case is classified as *Match* or *NonMatch*, but also why some comparisons remain *Inconclusive*.

4.2 Exploratory Analysis

Our exploratory analysis was designed to support the later modeling choices rather than to provide an exhaustive description of every variable. In particular, we focused on four questions: what kinds of missingness require preprocessing care, how the target classes are distributed, whether harder comparisons are more likely to produce inconclusive outcomes, and whether decision patterns vary across common forensic comparison categories.

Figure 1 summarizes the missing-value structure of the two model-ready datasets. The most notable pattern is that several variables exhibit structural missingness rather than random incompleteness. In KQ, the second-questioned-bullet quality fields are entirely missing, while in both KQ and QQ the intervening-bullets variables are missing for a substantial share of cases. This supports our decision to preserve the rows and handle missingness carefully during preprocessing instead of relying on indiscriminate row deletion.

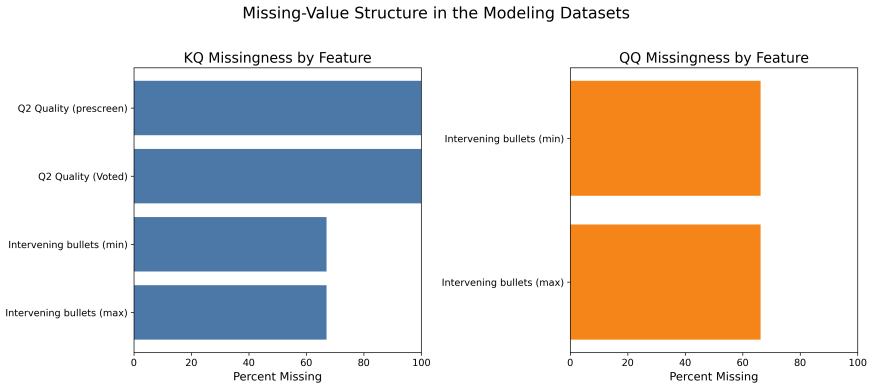


Figure 1: Missing-value structure in the KQ and QQ model-ready datasets.

Figure 2 shows the class distribution for the two datasets. Inconclusive is the largest class in both KQ and QQ, with 665 cases in KQ and 681 in QQ. This class imbalance reinforces the importance of treating the task as a multiclass problem and motivates our use of balanced accuracy as the primary evaluation metric.

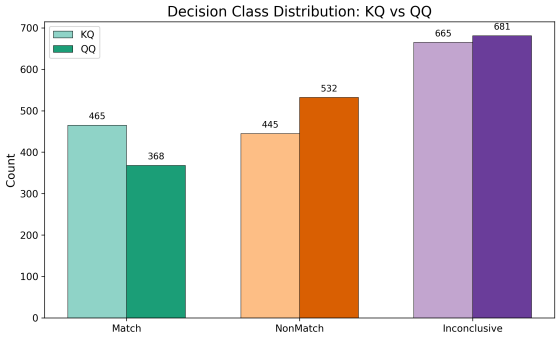


Figure 2: Decision class distribution for KQ and QQ.

Figure 3 examines how decision outcomes vary across difficulty levels. In both datasets, more difficult comparisons are associated with a larger share of *Inconclusive* outcomes, while easier comparisons are more likely to end in definitive *Match* or *NonMatch* decisions. This suggests that difficulty is not merely descriptive metadata, but a meaningful predictor of examiner uncertainty.

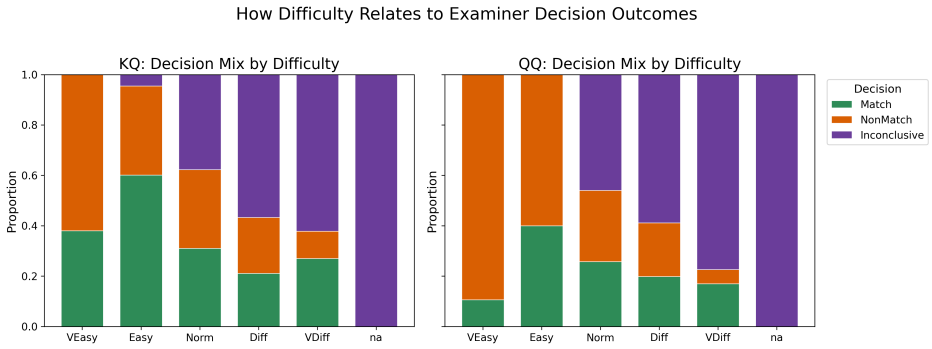


Figure 3: Decision proportions by difficulty level for KQ and QQ.

Finally, Figure 4 shows that decision mixtures differ across the most common detailed comparison categories. Some categories are dominated by definitive outcomes, whereas others contain much larger inconclusive shares. This supports the broader idea that firearm and ammunition context influences comparison difficulty and helps motivate later subgroup analysis and interpretation of feature-importance results.

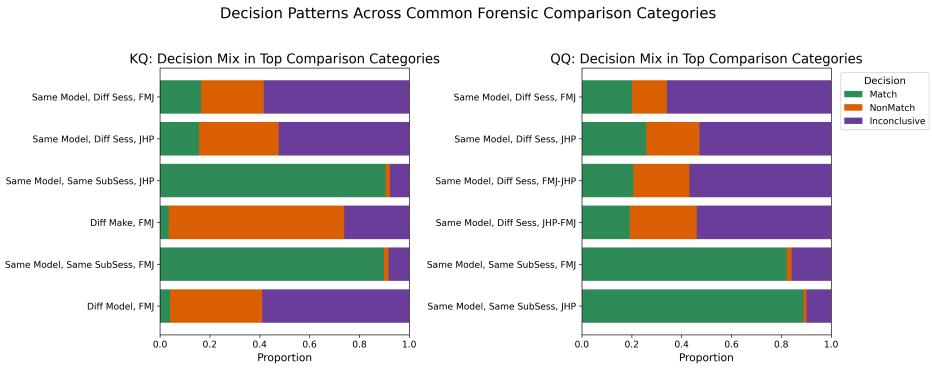


Figure 4: Decision mixtures across the most common detailed comparison categories in KQ and QQ.

Taken together, these exploratory findings justify the structure of our modeling pipeline. They show that the datasets require careful treatment of structural missingness, that the target variable is meaningfully imbalanced, and that both difficulty and comparison context are indicators of the examiner decision behavior.

5 Methods

We model the examiner's final decision as a classification problem with three possible outcomes: *Match*, *NonMatch*, and *Inconclusive*. Because the Known–Questioned (KQ) and Questioned–Questioned (QQ) datasets are structured slightly differently, we train separate models for each dataset. Before training the models, the decision labels and identifier columns are removed so that the remaining variables describe only the characteristics of the bullet comparisons and the conditions of the examination.

Before fitting the models, the data are cleaned and prepared for analysis. Missing numeric values are replaced using the median of each variable, and categorical variables are replaced with the most common category. These categorical variables are then converted into numeric form so that they can be used by the machine learning algorithms. After preprocessing, the data are split into training and testing sets using an 80/20 split while preserving the proportion of the three decision classes in both sets.

Random Forest is used as the main baseline model. Random Forest works by combining many decision trees, each trained on a different subset of the data, and then averaging their predictions. This approach is commonly used for

tabular data because it can capture nonlinear relationships between variables without requiring strong assumptions about the data. To reduce overfitting and improve model stability, the model parameters are tuned using five-fold cross-validation.

We also evaluate XGBoost, another tree-based ensemble method. Unlike Random Forest, which builds trees independently, XGBoost builds trees sequentially, where each new tree focuses on correcting errors made by earlier trees. This boosting process often leads to stronger predictive performance. For these models, categorical variables are encoded numerically and the models are trained using a fixed set of tuned parameters.

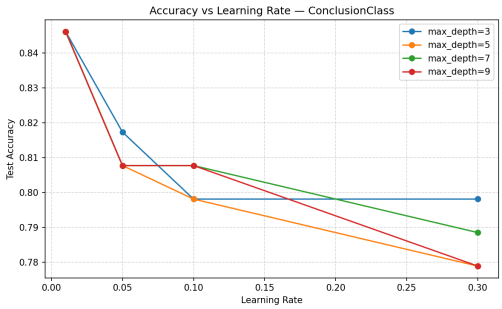
Although we also experimented with a reduced set of OBJECTIVE features that removed contextual variables, the main results reported here focus on the FULL feature models, which include both bullet-mark characteristics and comparison-condition variables.

Model performance is evaluated using several metrics. Accuracy measures the overall proportion of correct predictions, while balanced accuracy accounts for differences in class frequency. Confusion matrices are used to examine which decision categories are most often confused with each other. To better understand how the models make predictions, we also examine feature importance and SHAP explanations, which highlight which variables have the largest influence on the predicted outcomes.

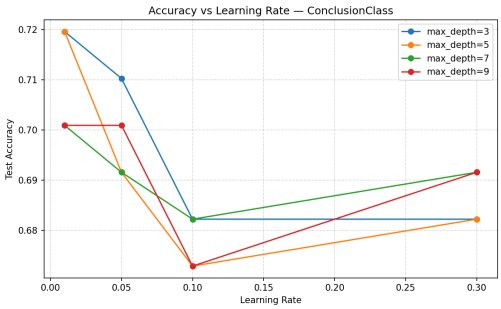
6 Results

The Random Forest models show clear predictive signal in both forensic settings when the FULL feature set is used. On the held-out test data, balanced accuracy is 0.681 for the KQ dataset and 0.691 for the QQ dataset, with overall accuracy around 0.68–0.69 in both cases. Since this is a three-class classification problem, these values are well above chance and suggest that the models are capturing real patterns in the examiner decision data.

To better understand how model parameters affect performance, we examined how test accuracy changes across different learning rates and tree depths for the XGBoost models. Figure 5 shows the results of this exploration for both the KQ and QQ datasets. For the KQ dataset (left), the highest accuracy occurs at smaller learning rates, particularly around 0.01, across several tree depths. Performance generally decreases slightly as the learning rate increases. For the QQ dataset (right), a similar pattern appears, with the best performance occurring at lower learning rates and moderate tree depths. These results guided the selection of the final model configuration used in our experiments. In general, smaller learning rates combined with moderate tree depths provided stable performance without overfitting.



(a) KQ dataset



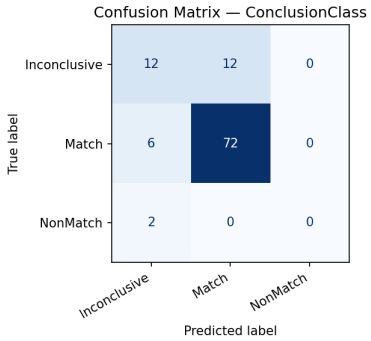
(b) QQ dataset

Figure 5: Test accuracy of XGBoost models across different learning rates and tree depths for the KQ and QQ datasets.

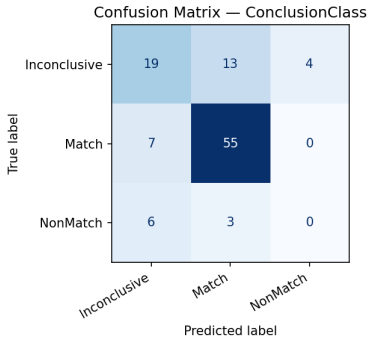
We also examine the confusion matrices for the XGBoost models shown in Figure 6. For the KQ dataset (left), most Match cases are predicted correctly, with 72 correct classifications. Some Inconclusive cases are predicted as Match, which suggests that borderline comparisons can be difficult for the model to separate. The few NonMatch cases in the test set are sometimes predicted as Inconclusive.

For the QQ dataset (right), the model correctly predicts 55 Match cases. However, there is again noticeable confusion between the Inconclusive and NonMatch categories. Some Inconclusive cases are predicted as Match, and several NonMatch cases are predicted as Inconclusive. This pattern is not surprising, since these categories often correspond to evidence that is close in strength in firearm comparisons.

Feature importance plots help explain which variables drive the predictions. The most influential features for both datasets are shown in Figure 7. In the KQ model (left), variables related to the examination process such as Phase,



(a) KQ dataset

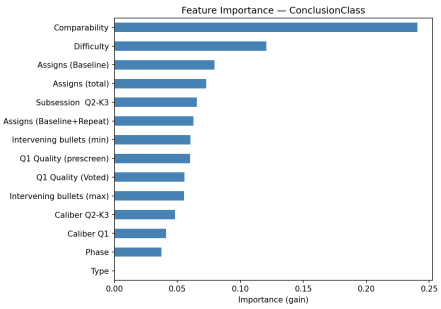


(b) QQ dataset

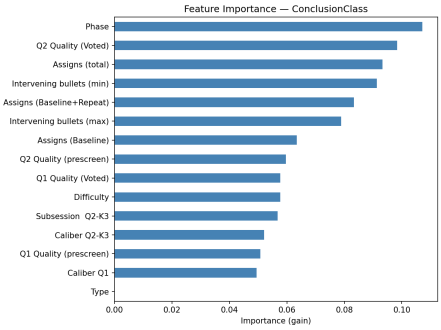
Figure 6: Confusion matrices for the XGBoost models predicting `ConclusionClass` using the FULL feature set.

Q2 Quality (Voted), and assignment-related measures appear among the strongest predictors. Measures related to intervening bullets also contribute substantially to the model predictions.

For the QQ model (right), Comparability and Difficulty appear as the most influential features, followed by assignment-related variables and intervening bullet measures. These results suggest that examiner decisions are influenced not only by bullet-mark characteristics but also by the broader comparison conditions and difficulty of the examination.



(a) KQ dataset



(b) QQ dataset

Figure 7: Feature importance for the XGBoost models trained on the FULL feature set.

In terms of predictive performance, the XGBoost model achieves strong results on the KQ dataset. For the three-class `ConclusionClass` prediction task, the model reaches a test accuracy of 0.8077, a macro ROC - AUC of 0.7644, and a cross-validation accuracy of 0.8266 ± 0.0393. For the QQ dataset, the XGBoost model achieves a test accuracy of 0.6916, a macro ROC–AUC of 0.8174, and a cross-validation accuracy of 0.7228 ± 0.0284.

Overall, these results show that machine learning models can capture meaningful patterns in firearm examination decisions. At the same time, the confusion matrices highlight the overlap between the NonMatch and Inconclusive categories, reflecting the inherent uncertainty present in many forensic comparisons.

7 Discussion and Conclusion

This study used interpretable machine learning models to examine examiner decision outcomes from the Bullet Black Box Study. By modeling firearm comparison decisions as a three-class prediction problem (*Match*, *NonMatch*, and *Inconclusive*), we were able to explore how different comparison features relate to examiner conclusions.The results show that the available features contain meaningful predictive signal. The Random Forest models achieved balanced accuracies of 0.681 for the KQ dataset and 0.691 for the QQ dataset, which are clearly above chance for a three-class classification task. The XGBoost models achieved similar or stronger performance, particularly for the KQ dataset. These results suggest that patterns in examiner decisions can be partially explained by the comparison characteristics and contextual variables included in the dataset.

At the same time, the confusion matrices highlight an important limitation: there is substantial overlap between the *NonMatch* and *Inconclusive* categories. The models frequently confuse these two outcomes in both datasets. This behavior reflects the nature of firearm comparison evidence rather than simply a modeling error. In practice, examiners may issue an inconclusive decision when the available markings are not strong enough to support a definitive exclusion. As a result, these two categories often represent evidence that is relatively close in strength.The feature-importance results provide additional insight into the factors that influence examiner decisions. Variables related to the comparison conditions, such as *Difficulty* and *Comparability*, appear among the most influential predictors in the QQ dataset. In the KQ dataset, examination-process variables such as *Phase* and quality assessments also play an important role. These findings suggest that examiner conclusions depend not only on bullet-mark characteristics but also on the broader context of the comparison task. More broadly, this work shows how machine learning can be used as a tool for studying forensic decision patterns. The goal is not to replace examiners with automated systems, but rather to better understand how different features and comparison conditions relate to the final decision outcomes. Interpretable models make it possible to identify which variables are most strongly associated with *Match*, *NonMatch*, and *Inconclusive* results.

There are several limitations to this analysis. The models rely only on the variables available in the public dataset and do not capture the full range of information that examiners may use during real casework. In addition, machine learning models can identify statistical patterns in the data but cannot fully represent the expert reasoning process used in forensic examinations.Despite these limitations, the study demonstrates that interpretable machine learning methods can provide useful insight into firearm comparison data. By analyzing how different variables relate to examiner outcomes, these methods help reveal the conditions under which comparisons become difficult and uncertainty arises in forensic decision-making.

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8 GITHUB LINK FOR PROJECT :

`https://github.com/panditk455/Machine-Learning-Project`